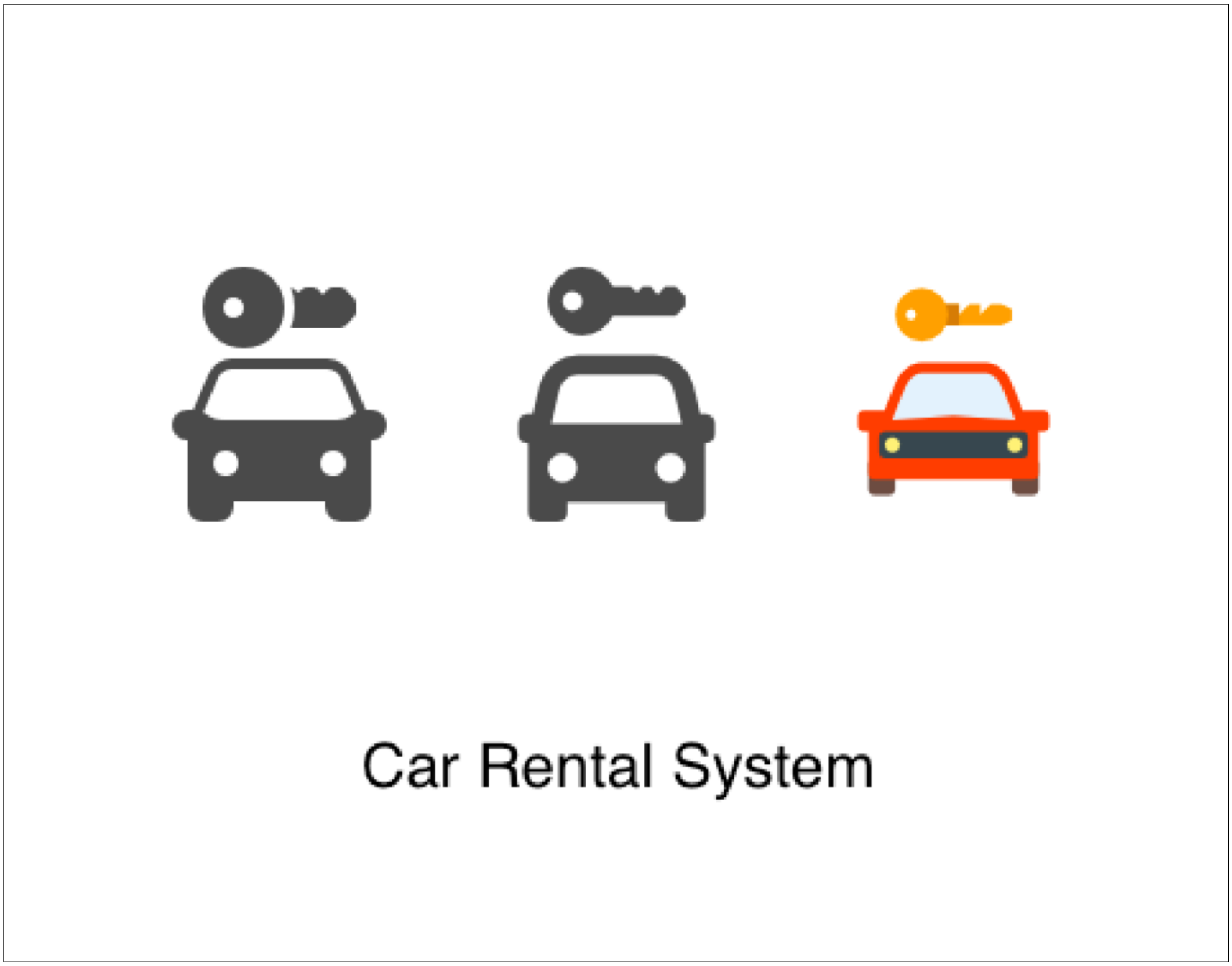

Database Design Project

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CONTENTS

DATA REQUIREMENTS

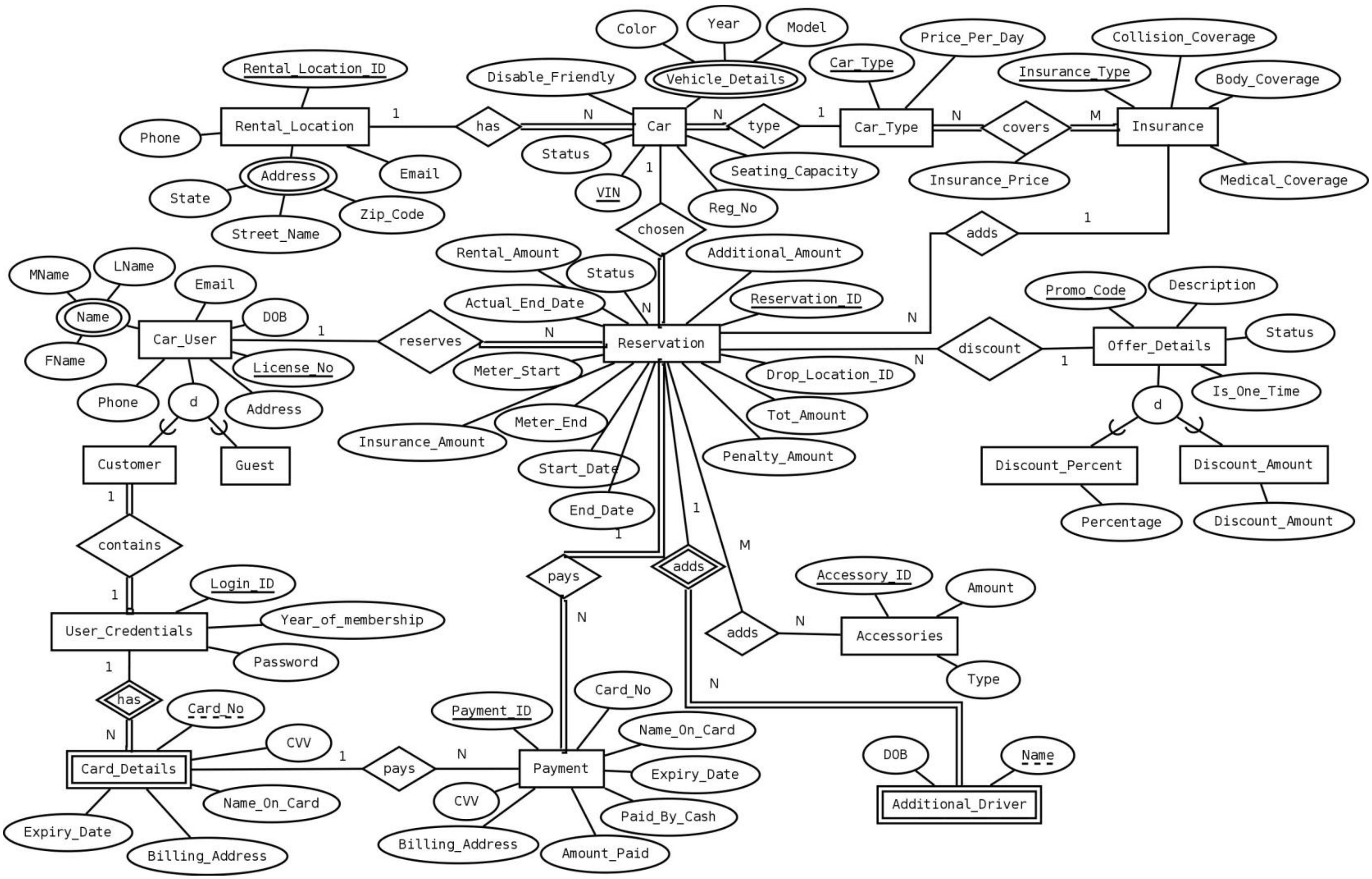
- The Car Rental Database needs to keep track of three main entities namely User, Car and Reservation.
- Car can be rented from a rental location with a specific address of each rental location. A rental location keeps track of contact phone number, contact email and rental location address with street name, state and zip code.

CAR RENTAL SYSTEM

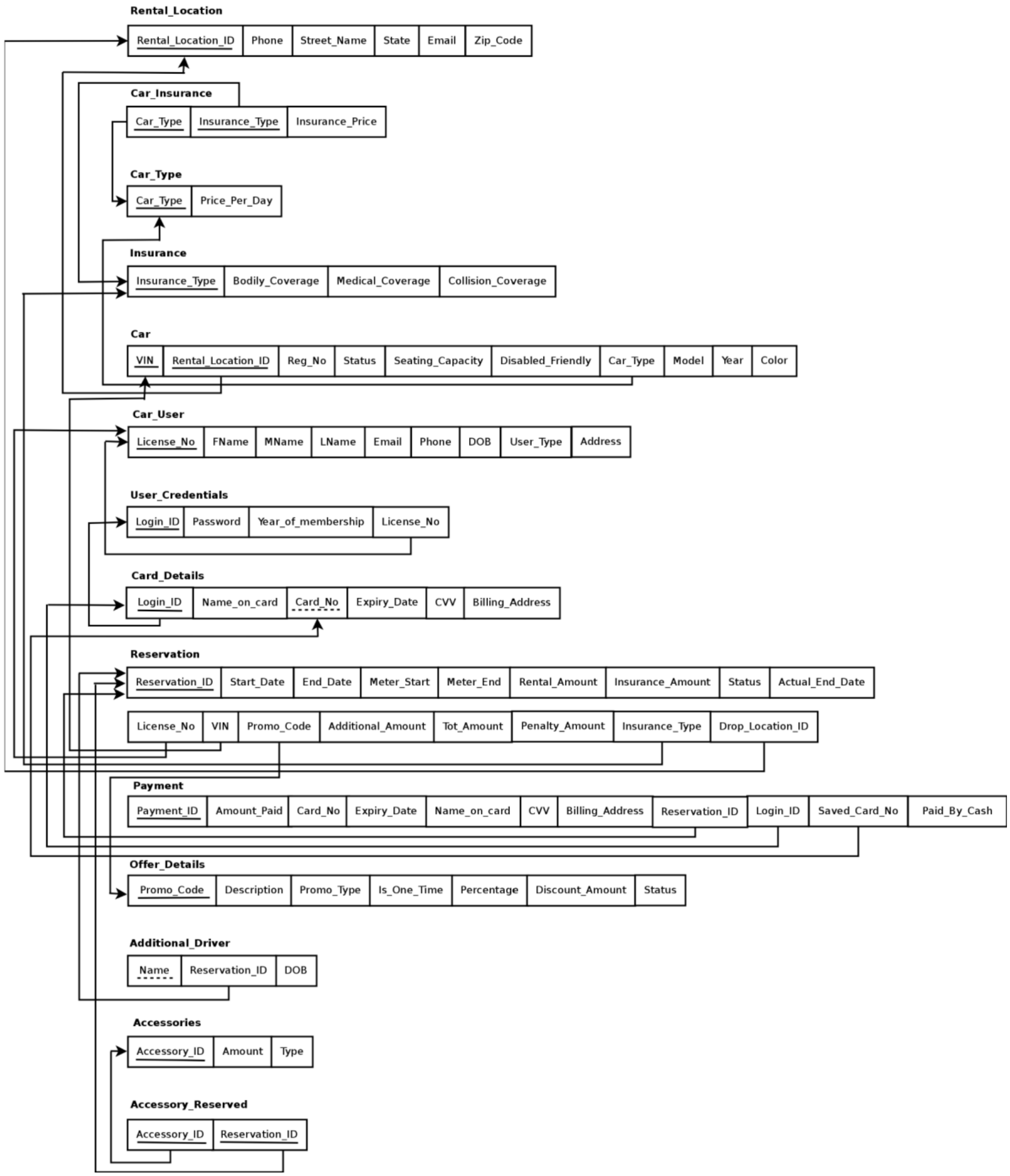
- A rental location has a number of cars for rental. Each car is described by VIN, Registration number, status as ‘Available’ and ‘Unavailable’, seating capacity of the car, whether it has disabled friendly feature, car color, car model and car manufactured year.
- Car has a parameter as Car Type. Car type can be ‘Economy’, ‘Standard’, ’SUV’, ‘Premium’ and ‘Minivan’. Car Type defines the rental price per day. A user can take Insurance per day for the rental car. There are different types of insurance each having different medical insurance coverage, collision coverage and car bodily coverage. Insurance Types are ‘Liability’ and ‘Comprehensive’. Car type and Insurance Type drives the Insurance price per day.
- A user can reserve a car for a number of days. He can use any valid promotional code which is maintained by status. When a user books a car he mentions the start date and end date for which he needs the car. The end date will be hypothetical at the time of reservation and updated with actual end date when the car is returned. If the Actual End Date is greater than End Date mentioned initially, then a penalty is imposed. The total amount is calculated based on start date, end date, rental price per day, insurance price per day and promotional code if any used.
- A user can cancel a reserved car before he/she rented the car. A reservation can have status as ‘Reserved’, ‘Completed’ and ‘Cancelled’. When the car is reserved, status will be in ‘Reserved’ Status. Once the car rental of the reservation is completed with due amount paid, the status will be ‘Completed’.
- A user can drop the car in any location. Car will be available for future use from the location where it is dropped.
- A user is categorized as guest and customer. User can continue reserving car as guest as long as he has not registered as customer. A user is uniquely identified by his/her license number. User information consists of his name as Fist name, middle name and last name, Email address, physical address, date of birth and contact number.
- A registered customer will be provided with login id and password. A customer can save his credit/ debit card details for future payment.

- Partial payment can be made at the time of reservation and the rest must be paid by the user during car return when actual end date is known. If user is a customer, he/she can pay through saved debit/ credit card details. A user irrespective of whether guest or customer has an option to pay by cash.
- A user can add any accessories as part of his/her reservation. Accessories can be ‘Car Seat’ and ‘GPS’. A user can add as many accessories he/she needs as part of the reservation.
- Additional driver can be added as part of his/her reservation. For each additional driver there will be an additional charges.

ENTITY RELATIONSHIP DIAGRAM



MAPPING



FUNCTIONAL DEPENDENCIES

- In Rental car location, Rental_Location_ID is the primary key

Rental_Location_ID —> {Phone, Email, Street_Name, State, Zip_Code}

- Type of the car defines the rental price of the car per day

  Car_Type —> Price_Per_Day

- Type of the insurance defines the insurance coverage

  Insurance_Type —> {Bodily_Coverage, Medical_Coverage, Collision_Coverage}

- Insurance Type and Car Type defines the Insurance price per day

  {Car_Type, Insurance_Type} —> {Insurance_Price}

- A user is defined by his/her License_No

  {License_No} —> {FName, Mname, Lname, Email, Address, Phone, DOB, User_Type}

- In a user credential, Login_ID defines the rest of the attributes in the entity

  {Login_ID} —> {Password, Year_Of_Membership, License_No}

- Login_ID and Card_No in Card_Details defines complete card information

  {Login_ID, Card_No} —> {Name_On_Card, Expiry_Date, CVV, Billing_Address}

- Reservation_ID drives all the other attributes in Reservation relation

  Reservation_ID —> {Start_Date, End_Date, Meter_Start, Meter_End, Rental_Amount, Insurance_Amount, Actual_End_Date, Status, License_No, VIN, Promo_Code, Additional_Amount, Tot_Amount, Insurance_Type, Penalty_Amount, Drop_Location_ID}

- Payment_ID is the primary key of Payment relation

  Payment_ID —> {Amount_Paid, Card_No, Expiry_Date, Name_On_Card, CVV, Billing_Address, Reservation_ID, Login_ID, Saved_Card_No, Paid_By_Cash}

- Promo_Code defines other attributes in Offer_Details relation

  Promo_Code —> {Description, Promo_Type, Is_One_Time, Percentage, Discounted_Amount, Status}

- Additional_Drivers relation

  {Reservation_ID,Name} —> DOB

- In Accessories relation, Accessory_ID drives the type and amount of the accessory

  Accessory_ID —> {Type, Amount}

<u>Note</u>: Since our mapping was already in 3NF, we did not decompose it further

SQL STATEMENTS

-- -- *Start of Coding*

-- --

-- *DROP STATEMENTS*

-- drop table RENTAL_LOCATION cascade constraints purge; drop table CAR_TYPE cascade constraints purge; drop table INSURANCE cascade constraints purge; drop table CAR_INSURANCE cascade constraints purge; drop table CAR_USER cascade constraints purge; drop table USER_CREDENTIALS cascade constraints purge; drop table CARD_DETAILS cascade constraints purge; drop table CAR cascade constraints purge; drop table RESERVATION cascade constraints purge; drop table PAYMENT cascade constraints purge; drop table OFFER_DETAILS cascade constraints purge; drop table ADDITIONAL_DRIVER cascade constraints purge; drop table ACCESSORIES cascade constraints purge; drop table ACCESSORY_RESERVED cascade constraints purge;

commit;

-- -- *CREATE TABLE STATEMENTS*

-- CREATE TABLE RENTAL_LOCATION

```
(
  Rental_Location_ID INT PRIMARY KEY,
  Phone CHAR(10) NOT NULL,
  Email VARCHAR(25),
  Street_Name VARCHAR(40) NOT NULL,
  State CHAR(2) NOT NULL,
  Zip_Code CHAR(6) NOT NULL
);
```

```
CREATE TABLE CAR_TYPE
(
  Car_Type VARCHAR(15) PRIMARY KEY,
  Price_Per_Day NUMBER(8,2) NOT NULL
);

CREATE TABLE INSURANCE
(
  Insurance_Type VARCHAR(15) PRIMARY KEY,
  Bodily_Coverage NUMBER(8,2) NOT NULL,
  Medical_Coverage NUMBER(8,2) NOT NULL,
  Collision_Coverage NUMBER(8,2) NOT NULL
);
CREATE TABLE CAR_INSURANCE
(
  Car_Type VARCHAR(15),
  Insurance_Type VARCHAR(15),
  Insurance_Price NUMBER(8,2) NOT NULL,
  PRIMARY KEY(Car_Type,Insurance_Type),
  CONSTRAINT CARTYPEFK
  FOREIGN KEY (Car_Type) REFERENCES CAR_TYPE(Car_Type)
        ON DELETE CASCADE,
  CONSTRAINT INSURANCETYPEFK   FOREIGN KEY (Insurance_Type) REFERENCES
INSURANCE(Insurance_Type)
        ON DELETE CASCADE
);

CREATE TABLE CAR_USER
```

```
(
  License_No VARCHAR(15) PRIMARY KEY,
  Fname VARCHAR(15) NOT NULL,
  Mname VARCHAR(1),
  Lname VARCHAR(15) NOT NULL,
  Email VARCHAR(25) NOT NULL UNIQUE,
  Address VARCHAR(100) NOT NULL,
  Phone CHAR(10) NOT NULL,
  DOB DATE NOT NULL,   User_Type
VARCHAR(10) NOT NULL
);
CREATE TABLE USER_CREDENTIALS
(
  Login_ID VARCHAR(15) PRIMARY KEY,
  Password VARCHAR(15) NOT NULL,
  Year_Of_Membership Char(4) NOT NULL ,
  License_No VARCHAR(15) NOT NULL,
  CONSTRAINT USRLIC
  FOREIGN KEY (License_No) REFERENCES CAR_USER(License_No)            ON DELETE
CASCADE
);

CREATE TABLE CARD_DETAILS
(
  Login_ID VARCHAR(15) NOT NULL,
  Name_On_Card VARCHAR(50) NOT NULL,
  Card_No CHAR(16) NOT NULL,
  Expiry_Date DATE NOT NULL,
```

```
  CVV CHAR(3) NOT NULL,
  Billing_Address VARCHAR(50) NOT NULL,
  PRIMARY KEY(Login_ID,Card_No),
  CONSTRAINT USRCARDFK
  FOREIGN KEY (Login_ID) REFERENCES USER_CREDENTIALS(Login_ID)        ON DELETE
CASCADE
);
CREATE TABLE CAR
(
  VIN CHAR(17) PRIMARY KEY,
  Rental_Location_ID INT NOT NULL,
  Reg_No VARCHAR(15) UNIQUE,
  Status VARCHAR(15) NOT NULL,
  Seating_Capacity INT NOT NULL,
  Disability_Friendly CHAR(1),
  Car_Type VARCHAR(15) NOT NULL,
  Model VARCHAR(20),
  Year CHAR(4),
  Color VARCHAR(10),
  CONSTRAINT CARVINTYPEFK   FOREIGN KEY (Car_Type) REFERENCES
CAR_TYPE(Car_Type)
        ON DELETE CASCADE,
  CONSTRAINT CARVINRENTALFK
  FOREIGN KEY (Rental_Location_ID) REFERENCES RENTAL_LOCATION(Rental_Location_ID)
        ON DELETE CASCADE
);

CREATE TABLE OFFER_DETAILS
```

```
(
  Promo_Code VARCHAR(15) PRIMARY KEY,
  Description VARCHAR(50),   Promo_Type
VARCHAR(20) NOT NULL,
  Is_One_Time CHAR(1),
  Percentage DECIMAL(5,2),
  Discounted_Amount NUMBER(8,2),
  Status VARCHAR(10) NOT NULL);
CREATE TABLE RESERVATION
(
  Reservation_ID INT PRIMARY KEY,
  Start_Date DATE NOT NULL,
  End_Date DATE NOT NULL,
  Meter_Start INT NOT NULL,
  Meter_End INT,
  Rental_Amount NUMBER(8,2) NOT NULL,
  Insurance_Amount NUMBER(8,2) NOT NULL,
  Actual_End_Date DATE NULL,
  Status VARCHAR(10) NOT NULL,
  License_No VARCHAR(15) NOT NULL,
  VIN CHAR(17) NOT NULL,
  Promo_Code VARCHAR(15),
  Additional_Amount NUMBER(8,2),
  Tot_Amount NUMBER(8,2) NOT NULL,
  Insurance_Type VARCHAR(15),
  Penalty_Amount NUMBER(8,2),
  Drop_Location_ID INT,
  CONSTRAINT RSERVLOCATIONFK
```

```
 FOREIGN KEY (Drop_Location_ID) REFERENCES RENTAL_LOCATION(Rental_Location_ID)
        ON DELETE CASCADE,
 CONSTRAINT RESLICENSEFK
 FOREIGN KEY (License_No) REFERENCES CAR_USER(License_No)
        ON DELETE CASCADE,
 CONSTRAINT VINRESERVATIONFK
 FOREIGN KEY (VIN) REFERENCES CAR(VIN)
        ON DELETE CASCADE,
 CONSTRAINT PROMORESERVATIONFK
 FOREIGN KEY (Promo_Code) REFERENCES OFFER_DETAILS(Promo_Code)
        ON DELETE CASCADE,
 CONSTRAINT INSURESERVATIONFK   FOREIGN KEY (Insurance_Type) REFERENCES
INSURANCE(Insurance_Type)
        ON DELETE CASCADE
);

CREATE TABLE PAYMENT
(
 Payment_ID INT PRIMARY KEY,
 Amount_Paid NUMBER(8,2) NOT NULL,
 Card_No CHAR(16),
 Expiry_Date DATE,   Name_On_Card
VARCHAR(50),
 CVV CHAR(3),
 Billing_Address VARCHAR(50),
 Reservation_ID INT NOT NULL,
 Login_ID VARCHAR(15),
 Saved_Card_No CHAR(16),
```

```
 Paid_By_Cash CHAR(1),
 CONSTRAINT PAYMENTRESERVATIONFK   FOREIGN KEY (Reservation_ID) REFERENCES
RESERVATION(Reservation_ID)
        ON DELETE CASCADE,
 CONSTRAINT PAYMENTLOGINFK
 FOREIGN KEY (Login_ID,Saved_Card_No) REFERENCES CARD_DETAILS(Login_ID,Card_No)
        ON DELETE CASCADE
);
CREATE TABLE ADDITIONAL_DRIVER
(
 Reservation_ID INT,
 NAME VARCHAR(50) NOT NULL,
 DOB DATE NOT NULL,
 PRIMARY KEY(Reservation_ID,NAME),
 CONSTRAINT ADDTIONALFK
 FOREIGN KEY (Reservation_ID) REFERENCES RESERVATION(Reservation_ID)
        ON DELETE CASCADE
);

CREATE TABLE ACCESSORIES
(
 Accessory_ID INT PRIMARY KEY,
 Type VARCHAR(15) NOT NULL,
 Amount NUMBER(8,2) NOT NULL
);

CREATE TABLE ACCESSORY_RESERVED
(
```

```
 Accessory_ID INT,
 Reservation_ID INT,
 PRIMARY KEY(Accessory_ID,Reservation_ID),
 CONSTRAINT ACCESSORYRESERVFK   FOREIGN KEY (Reservation_ID) REFERENCES
RESERVATION(Reservation_ID)
        ON DELETE CASCADE,
 CONSTRAINT ACCESSFK
 FOREIGN KEY (Accessory_ID) REFERENCES ACCESSORIES(Accessory_ID)
        ON DELETE CASCADE);commit; -------------------------------------------------
```

-- *INSERT STATEMENTS*

```
-------------------------------------------------
INSERT ALL
INTO RENTAL_LOCATION
(Rental_Location_ID,Phone,Email,Street_Name,State,Zip_Code)
VALUES
(101,'9726031111','adams12@gmail.com','980 Addison Road, Dallas','TX',75123)
INTO RENTAL_LOCATION
(Rental_Location_ID,Phone,Email,Street_Name,State,Zip_Code)
VALUES
(102,'9726032222','bobw@gmail.com',' 111, Berlington Road, Dallas','TX',75243)
INTO RENTAL_LOCATION
(Rental_Location_ID,Phone,Email,Street_Name,State,Zip_Code)
VALUES
(103,'9721903121','patric.clever@gmail.com',' 9855 Shadow Way, Dallas','TX',
75211)
INTO RENTAL_LOCATION
(Rental_Location_ID,Phone,Email,Street_Name,State,Zip_Code)
VALUES
```

```
(104,'721903121',NULL,'434 Harrodswood Road, Irving','TX',76512)
INTO RENTAL_LOCATION
(Rental_Location_ID,Phone,Email,Street_Name,State,Zip_Code)
VALUES
(105,'5026981045','julier@gmail.com','7788 internal Drive, Irving','TX',77888)
SELECT * FROM Dual;
INSERT ALL
INTO CAR_TYPE
(Car_Type,Price_Per_Day)
VALUES
('Economy',19.95)
INTO CAR_TYPE
(Car_Type,Price_Per_Day)
VALUES
('Standard',29.95)
INTO CAR_TYPE
(Car_Type,Price_Per_Day)
VALUES
('SUV',89.95)
INTO CAR_TYPE
(Car_Type,Price_Per_Day)
VALUES
('MiniVan',109.95)
INTO CAR_TYPE
(Car_Type,Price_Per_Day)
VALUES
('Premium',149.95)
SELECT * FROM dual;
```

```
INSERT ALL
INTO INSURANCE
(Insurance_Type,Bodily_Coverage,Medical_Coverage,Collision_Coverage)
VALUES
('Liability',25000.00,50000.00,0.00)
INTO INSURANCE (Insurance_Type,Bodily_Coverage,Medical_Coverage,Collision_Coverage)
VALUES
('Comprehensive',50000.00,50000.00,50000.00)
SELECT * FROM dual;

INSERT ALL
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('Economy','Liability',9.99)
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('Standard','Liability',10.99)
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('SUV','Liability',12.99)
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('MiniVan','Liability',14.99)
INTO CAR_INSURANCE
```

```
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('Premium','Liability',19.99)
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('Economy','Comprehensive',19.99)
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('Standard','Comprehensive',19.99)
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('SUV','Comprehensive',24.99)
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('MiniVan','Comprehensive',29.99)
INTO CAR_INSURANCE
(Car_Type,Insurance_Type,Insurance_Price)
VALUES
('Premium','Comprehensive',49.99)
SELECT * FROM dual;

INSERT ALL
INTO CAR_USER
(License_No,FName,MName,Lname,Email,Address,Phone,DOB,USER_TYPE)
```

VALUES

('E12905109','Patrick','G','Cleaver','patric.c@yahoo.com','1701 N.Campbell Rd, Dallas, TX-75243','5022196058',TO_DATE('1970/01/10', 'yyyy/mm/dd'),'Guest')

INTO CAR_USER

(License_No,FNAME,MNAME,LNAME,Email,Address,Phone,DOB,USER_TYPE)

VALUES

('C11609103','Courtney',NULL,'Rollins','courtney.r@hotmail.com','1530 S.Campbell Rd','4697891045',TO_DATE('1990/03/20', 'yyyy/mm/dd'),'Customer')

INTO CAR_USER

(License_No,FNAME,MNAME,LNAME,Email,Address,Phone,DOB,USER_TYPE)

VALUES

('G30921561','Glenn',NULL,'Tucker','glenn.t@hotmail.com','101 Meritline drive','8590125607',TO_DATE('1964/11/11', 'yyyy/mm/dd'),'Customer')

INTO CAR_USER

(License_No,FNAME,MNAME,LNAME,Email,Address,Phone,DOB,USER_TYPE)

VALUES

('R12098127','Ron',NULL,'Harper','ron.harper@hotmail.com','43 Greenville Road','2048015647',TO_DATE('1987/04/24', 'yyyy/mm/dd'),'Guest')

INTO CAR_USER

(License_No,FNAME,MNAME,LNAME,Email,Address,Phone,DOB,USER_TYPE)

VALUES

('M12098127','Manoj',NULL,'Punwani','manoj123@gmail.com','43 Greenville Road','2048015647',TO_DATE('1987/04/24', 'yyyy/mm/dd'),'Customer')

SELECT * FROM dual;

INSERT ALL

INTO USER_CREDENTIALS

(Login_ID,Password,Year_Of_Membership,License_No)

VALUES

('courtney90','bc125ac','2009','C11609103')

```
INTO USER_CREDENTIALS
(Login_ID,Password,Year_Of_Membership,License_No)
VALUES
('glenn64','macpro99','2011','G30921561')
INTO USER_CREDENTIALS
(Login_ID,Password,Year_Of_Membership,License_No)
VALUES
('manoj87','windows99','2008','M12098127')
SELECT * FROM dual;

INSERT ALL
INTO CARD_DETAILS
(Login_ID,Name_On_Card,Card_No,Expiry_Date,CVV,Billing_Address)
VALUES
('courtney90','Courtney Rollins','4735111122223333',TO_DATE('2018/01/15', 'yyyy/ mm/dd'),'833','1530 S.Campbell Rd, Dallas, TX 75251')
INTO CARD_DETAILS
(Login_ID,Name_On_Card,Card_No,Expiry_Date,CVV,Billing_Address)
VALUES
('manoj87','Manoj Punwani','4233908110921001',TO_DATE('2019/12/31', 'yyyy/mm/ dd'),'419','9855 Shadow Way, TX 75243')
SELECT * FROM dual;

INSERT ALL
INTO CAR
(VIN,Rental_Location_ID,Reg_No,Status,Seating_Capacity,Disability_Friendly,Car_Ty pe,Model,Year,Color)
VALUES
('F152206785240289',101,'TXF101','Available',
5,'N','Economy','Mazda3','2007','Gold')
```

```
INTO CAR
(VIN,Rental_Location_ID,Reg_No,Status,Seating_Capacity,Disability_Friendly,Car_Ty pe,Model,Year,Color)
VALUES
('T201534710589051',101,'KYQ101','Available',5,'Y','Standard','Toyota
Camry','2012','Grey')
INTO CAR
(VIN,Rental_Location_ID,Reg_No,Status,Seating_Capacity,Disability_Friendly,Car_Ty pe,Model,Year,Color)
VALUES
('E902103289341098',102,'XYZ671','Available',
5,NULL,'Premium','BMW','2015','Black')
INTO CAR
(VIN,Rental_Location_ID,Reg_No,Status,Seating_Capacity,Disability_Friendly,Car_Ty pe,Model,Year,Color)
VALUES
('R908891209418173',103,'DOP391','Unavailable',7,NULL,'SUV','Acura
MDX','2014','White')
INTO CAR
(VIN,Rental_Location_ID,Reg_No,Status,Seating_Capacity,Disability_Friendly,Car_Ty pe,Model,Year,Color)
VALUES
('N892993994858292',104,'RAC829','Available',
15,NULL,'MiniVan','Sienna','2013','Black')
SELECT * FROM dual;

INSERT ALL
INTO OFFER_DETAILS
(PROMO_CODE,DESCRIPTION,PROMO_TYPE,IS_ONE_TIME,PERCENTAGE,DISCOUNTED_AMOUNT,Statu s)
VALUES
('CHRISTMAS10','Christmas 10% offer','Percentage','N',10.00,NULL,'Available')
INTO OFFER_DETAILS
(PROMO_CODE,DESCRIPTION,PROMO_TYPE,IS_ONE_TIME,PERCENTAGE,DISCOUNTED_AMOUNT,Statu s)
VALUES
```

('July25','July $25.00 discount','Discounted Amount','Y',NULL,25.00,'Expired')

INTO OFFER_DETAILS

(PROMO_CODE,DESCRIPTION,PROMO_TYPE,IS_ONE_TIME,PERCENTAGE,DISCOUNTED_AMOUNT,Statu s)

VALUES

('LaborDay5','Labor Day $5.00 offer','Discounted Amount','Y',NULL,5.00,'Expired') INTO OFFER_DETAILS

(PROMO_CODE,DESCRIPTION,PROMO_TYPE,IS_ONE_TIME,PERCENTAGE,DISCOUNTED_AMOUNT,Statu s)

VALUES

('NewYear10','New Year 10% offer','Percentage','N',10.00,NULL,'Available')

INTO OFFER_DETAILS

(PROMO_CODE,DESCRIPTION,PROMO_TYPE,IS_ONE_TIME,PERCENTAGE,DISCOUNTED_AMOUNT,Statu s)

VALUES

('VeteranDay15','New Year 15% offer','Percentage','N',15.00,NULL,'Expired')

SELECT * FROM dual;

INSERT ALL

INTO RESERVATION

(Reservation_ID,Start_Date,End_Date,Meter_Start,Meter_End,Rental_Amount,Insurance _Amount,Status,Actual_End_Date,License_No,VIN,Promo_Code,Additional_Amount,Tot_Am ount,Penalty_Amount,Insurance_Type,Drop_Location_ID)

VALUES

(1,TO_DATE('2015/11/06', 'yyyy/mm/dd'),TO_DATE('2015/11/12', 'yyyy/mm/dd'), 81256,81300,119.70,9.95,'Completed',TO_DATE('2015/11/12', 'yyyy/mm/ dd'),'E12905109','F152206785240289',NULL,NULL,129.65,0.00,'Liability',101)

INTO RESERVATION

(Reservation_ID,Start_Date,End_Date,Meter_Start,Meter_End,Rental_Amount,Insurance _Amount,Status,Actual_End_Date,License_No,VIN,Promo_Code,Additional_Amount,Tot_Am ount,Penalty_Amount,Insurance_Type,Drop_Location_ID)

VALUES

(2,TO_DATE('2015/10/20', 'yyyy/mm/dd'),TO_DATE('2015/10/24', 'yyyy/mm/dd'), 76524,76590,119.80,9.95,'Completed',TO_DATE('2015/10/24', 'yyyy/mm/ dd'),'C11609103','T201534710589051',NULL,NULL,129.75,0.00,'Liability',101)

```
INTO RESERVATION

(Reservation_ID,Start_Date,End_Date,Meter_Start,Meter_End,Rental_Amount,Insurance
_Amount,Status,Actual_End_Date,License_No,VIN,Promo_Code,Additional_Amount,Tot_Am
ount,Penalty_Amount,Insurance_Type,Drop_Location_ID)

VALUES

(3,TO_DATE('2015/12/06', 'yyyy/mm/dd'),TO_DATE('2015/12/12', 'yyyy/mm/dd'), 82001,NULL,
659.40,29.95,'Reserved',NULL,'C11609103','N892993994858292','NewYear10',NULL,
689.35,0.00,'Comprehensive',104)

INTO RESERVATION

(Reservation_ID,Start_Date,End_Date,Meter_Start,Meter_End,Rental_Amount,Insurance
_Amount,Status,Actual_End_Date,License_No,VIN,Promo_Code,Additional_Amount,Tot_Am
ount,Penalty_Amount,Insurance_Type,Drop_Location_ID)

VALUES

(4,TO_DATE('2015/09/01', 'yyyy/mm/dd'),TO_DATE('2015/09/02', 'yyyy/mm/dd'),
51000,51100,89.95,24.95,'Completed',TO_DATE('2015/09/02', 'yyyy/mm/
dd'),'C11609103','R908891209418173',NULL,NULL,114.90,0.00,'Comprehensive',103)

INTO RESERVATION

(Reservation_ID,Start_Date,End_Date,Meter_Start,Meter_End,Rental_Amount,Insurance
_Amount,Status,Actual_End_Date,License_No,VIN,Promo_Code,Additional_Amount,Tot_Am
ount,Penalty_Amount,Insurance_Type,Drop_Location_ID)

VALUES

(5,TO_DATE('2015/08/13', 'yyyy/mm/dd'),TO_DATE('2015/08/15', 'yyyy/mm/dd'),
51000,51100,299.00,99.9,'Completed',TO_DATE('2015/08/15', 'yyyy/mm/
dd'),'R12098127','E902103289341098',NULL,NULL,398.90,0.00,'Comprehensive',105)

SELECT * FROM dual;


INSERT ALL

INTO PAYMENT

(Payment_ID,Amount_Paid,Card_NO,Expiry_Date,Name_On_Card,CVV,Billing_Address,Rese
rvation_ID,Login_ID,Saved_Card_No,Paid_By_Cash)

VALUES

(1001,129.65,'4735111122223333',TO_DATE('2018/01/15', 'yyyy/mm/dd'),'Patric Clever','100','1530 S.Campbell Rd,
Dallas, TX 75251',1,NULL,NULL,NULL)
```

```
INTO PAYMENT
(Payment_ID,Amount_Paid,Card_NO,Expiry_Date,Name_On_Card,CVV,Billing_Address,Rese
rvation_ID,Login_ID,Saved_Card_No,Paid_By_Cash)
VALUES
(1002,300.00,NULL,NULL,NULL,NULL,NULL,5,NULL,NULL,'Y')
INTO PAYMENT (Payment_ID,Amount_Paid,Card_NO,Expiry_Date,Name_On_Card,CVV,Billing_Address,Rese
rvation_ID,Login_ID,Saved_Card_No,Paid_By_Cash)
VALUES
(1003,98.90,NULL,NULL,NULL,NULL,NULL,5,NULL,NULL,'Y')
INTO PAYMENT
(Payment_ID,Amount_Paid,Card_NO,Expiry_Date,Name_On_Card,CVV,Billing_Address,Rese
rvation_ID,Login_ID,Saved_Card_No,Paid_By_Cash)
VALUES
(1004,689.35,NULL,NULL,NULL,NULL,NULL,3,'courtney90','4735111122223333',NULL)
INTO PAYMENT
(Payment_ID,Amount_Paid,Card_NO,Expiry_Date,Name_On_Card,CVV,Billing_Address,Rese
rvation_ID,Login_ID,Saved_Card_No,Paid_By_Cash)
VALUES
(1005,114.91,NULL,NULL,NULL,NULL,NULL,4,NULL,NULL,'Y')
SELECT * FROM dual;

INSERT ALL
INTO ADDITIONAL_DRIVER
(Reservation_ID,Name,DOB)
VALUES
(1,'William Smith',TO_DATE('1970/07/15', 'yyyy/mm/dd'))
INTO ADDITIONAL_DRIVER
(Reservation_ID,Name,DOB)
VALUES
(2,'Green Taylor',TO_DATE('1987/06/15', 'yyyy/mm/dd'))
```

```
INTO ADDITIONAL_DRIVER
(Reservation_ID,Name,DOB)
VALUES
(2,'Robert Moore',TO_DATE('1990/12/17', 'yyyy/mm/dd'))
INTO ADDITIONAL_DRIVER
(Reservation_ID,Name,DOB)
VALUES
(4,'Brad Cook',TO_DATE('1966/12/12', 'yyyy/mm/dd'))
INTO ADDITIONAL_DRIVER
(Reservation_ID,Name,DOB)
VALUES
(5,'Steve Fouts',TO_DATE('1976/05/28', 'yyyy/mm/dd'))
SELECT * FROM Dual;

INSERT ALL
INTO ACCESSORIES
(Accessory_ID,Type,Amount) VALUES
(1,'GPS Navigation',49.95) INTO ACCESSORIES
(Accessory_ID,Type,Amount) VALUES
(2,'GPS Navigation',49.95) INTO ACCESSORIES
(Accessory_ID,Type,Amount) VALUES
(3,'GPS Navigation',49.95) INTO ACCESSORIES
(Accessory_ID,Type,Amount)
VALUES
(4,'Baby Seater',29.95)
INTO ACCESSORIES
(Accessory_ID,Type,Amount)
VALUES
```

```
(5,'Baby Seater',29.95)
SELECT * FROM dual;

INSERT ALL
INTO ACCESSORY_RESERVED
(Accessory_ID,Reservation_ID)
VALUES
(1,1)
INTO ACCESSORY_RESERVED
(Accessory_ID,Reservation_ID)
VALUES
(1,4)
INTO ACCESSORY_RESERVED
(Accessory_ID,Reservation_ID)
VALUES
(5,5)
INTO ACCESSORY_RESERVED
(Accessory_ID,Reservation_ID)
VALUES
(5,2)
INTO ACCESSORY_RESERVED
(Accessory_ID,Reservation_ID)
VALUES
(2,4) SELECT * FROM dual;
commit; -------------------------
------------------------ --
```

PL/SQL

```
-------------------------------------------------
```

/*

1. UpdateStatus

This procedure is called as daily batch process (preferably a nightly automatic batch process). The purpose of this procedure is, when the user has paid the total amount and credit is realized by respective credit card bank, following are the updates that need to happen. The reason it is better to be a nightly batch process is, usually a credit card transaction gets confirmed in mid-night and once it is confirmed, the application can insert a payment record based on the web service communication with credit card agency. Once Payment record is created in our car rental system following actions can be taken as part of the procedure.

a. Update the status of the Reservation record as 'Completed'

b. Update the Status of the car as 'Available'

c. Update the Rental_Location_ID of Car to the Drop_Location_ID of the reservation as the car will be available going forward from where it is dropped

d. Update the status of the promo_code if used, to 'Expired', if it is one time use promo_code

*/

```
-- 1. Update Reservation, Promo_Code and Car status with Drop Location ID

CREATE OR REPLACE PROCEDURE UpdateStatus AS

BEGIN DECLARE

  thisReservation RESERVATION%ROWTYPE; CURSOR

reservationCursor IS

    SELECT R.* FROM RESERVATION R WHERE R.STATUS = 'Reserved' AND

    R.TOT_AMOUNT <= (SELECT SUM(AMOUNT_PAID) FROM PAYMENT P WHERE
P.RESERVATION_ID = R.RESERVATION_ID)

    FOR UPDATE OF STATUS; BEGIN

  OPEN reservationCursor;

LOOP

  FETCH reservationCursor INTO thisReservation;

   EXIT WHEN (reservationCursor%NOTFOUND);

  -- Update Reservation Status

  UPDATE RESERVATION SET STATUS = 'Completed'

  WHERE CURRENT OF reservationCursor;
```

```
  -- Update Promo_Code Status
  UPDATE OFFER_DETAILS SET STATUS = 'Expired'
  WHERE PROMO_CODE = thisReservation.PROMO_CODE AND IS_ONE_TIME = 'Y';
  -- Update Rental_Location Of Car as Drop Location ID and Car Status
  UPDATE CAR SET RENTAL_LOCATION_ID = thisReservation.Drop_Location_ID,STATUS =
'Available'
  WHERE CAR.VIN = thisReservation.VIN;
END LOOP;
CLOSE reservationCursor;
END;
END UpdateStatus;

-- Call the procedure
SET SERVEROUTPUT ON
BEGIN
UpdateStatus;
END;
```

/* ***2.***

IncreaseInsuranceRate

This procedure updates the insurance price of each car based on the percentage that is passed to the procedure. This procedure can be referred by the application, whenever there is a rise in car insurance rate */

```
-- 2. Increase the insurance price by given percentage

CREATE OR REPLACE PROCEDURE IncreaseInsuranceRate(percentage IN NUMBER) AS

BEGIN DECLARE

  thisCarInsurance CAR_Insurance%ROWTYPE; CURSOR

carInsuranceCursor IS

    SELECT * FROM CAR_INSURANCE

    FOR UPDATE OF Insurance_Price;

BEGIN

  OPEN carInsuranceCursor;

LOOP

  FETCH carInsuranceCursor INTO thisCarInsurance;

   EXIT WHEN (carInsuranceCursor%NOTFOUND);


  UPDATE CAR_INSURANCE SET Insurance_Price = (Insurance_Price +
(Insurance_Price*percentage/100))

  WHERE CURRENT OF carInsuranceCursor;

END LOOP;

CLOSE carInsuranceCursor;

END;

END IncreaseInsuranceRate;
```

Print Reservations

This PL/SQL prints the list of car that are reserved by the user, past end date and not yet returned */

/* ***3.***

-- 3. Print Reservations whose End Date is in the past and Car not yet returned

SET SERVEROUTPUT ON DECLARE

thisVIN CAR.VIN%TYPE;

thisLicenseNo CAR_USER.License_No%TYPE; thisCarUser

CAR_USER%ROWTYPE; CURSOR openReservationCursor IS

SELECT VIN,License_No FROM RESERVATION WHERE STATUS = 'Reserved' AND END_DATE
< SYSDATE AND ACTUAL_END_DATE IS NULL;

BEGIN

OPEN openReservationCursor;

LOOP

FETCH openReservationCursor INTO thisVIN,thisLicenseNo;

EXIT WHEN (openReservationCursor%NOTFOUND); SELECT * INTO thisCarUser FROM CAR_USER WHERE

License_No = thisLicenseNo;

dbms_output.put_line('User ' || thisCarUser.Fname || ' ' || thisCarUser.Mname || ' ' || thisCarUser.Lname || ' ' ||
'whose email id is ' || thisCarUser.Email
||

' has not returned the car with VIN ' || thisVIN ||'.');

END LOOP;

CLOSE openReservationCursor;

END;

<u>Print Promo Code</u>

This PL/SQL prints the list of promo_codes which are NOT one time use, and the number of times those promo_codes are used

*/

-- 4. Print Promo code which are NOT of type One Time and number of times it is used till now. SET SERVEROUTPUT

ON DECLARE

/* ***4.***

```
 thispromoCode                    OFFER_DETAILS.Promo_Code%TYPE;
thispromoCodeCount          INT;          --                  thisLicenseNo
CAR_USER.License_No%TYPE;
--  thisCarUser CAR_USER%ROWTYPE;
 CURSOR openOfferDetailsCursor IS     SELECT Promo_Code FROM OFFER_DETAILS WHERE
Is_One_Time <> 'Y';
BEGIN
 OPEN openOfferDetailsCursor;
LOOP
 FETCH openOfferDetailsCursor INTO thispromoCode;
  EXIT WHEN (openOfferDetailsCursor%NOTFOUND);
 SELECT COUNT(*) INTO thispromoCodeCount FROM RESERVATION WHERE PROMO_CODE = thispromoCode;
 dbms_output.put_line( 'Promo Code ' || thispromoCode || ' has been used ' || thispromoCodeCount || ' times.');
END LOOP;
 CLOSE openOfferDetailsCursor;
END;
------------------------------------------------
-- End of Coding
------------------------------------------------
```